Curriculum vitae

Personal details and the date of the CV

Surname: FUNG

First name: Pak Lun

Preferred name: Alan

Residence: Finland

Date of the CV: September 2026

Research profile: https://orcid.org/0000-0003-3493-1383

Education

9/2023−12/2025 Master of Business Administration,
Specialized in **Managing Sustainability and Systems Change**,
Lapland University of Applied Science
Master thesis: Challenges and Insights from Pioneer Higher Education Institutions Utilising Carbon Roadmap

3/2019−2/2022 Doctoral of Science, major in **Atmospheric Science,**
Institute for Atmospheric and Earth System Research (INAR/Physics),
University of Helsinki
Doctoral dissertation (Pass with distinction): Derivation of Black Carbon Proxies in an Integrated Urban Air Quality Monitoring Network
Fields: Urban aerosols, air pollutant proxy, air quality

8/2015−8/2018 Master of Science, major in **Atmospheric Science**, **University of Helsinki**
Master thesis: Ozone deposition over a boreal lake by the eddy covariance method

(1/2016−6/2016) Exchange study in the department of Earth and Ecosystem Science,
Lund University
Satellite Remote Sensing (Pass with distinction)

Work experience

6/2026−Present Postdoctoral Researcher in **Finnish Meteorological Institute**

- Developed machine learning models to identify urban heat island hotspots at Helsinki Metropolitan Area under the Climate Impacts and Adaptation group

4/2025−2/2026 Air Quality Expert/Data Analyst in **MegaSense Oy**

- Performed data analytics of urban traffic and air quality modeling to support the implementation of green navigation application by using multiple sources of geospatial data

5/2018−12/2024 (Post-)Doctoral researcher in **University of Helsinki**

- Conducted spatial analysis and modelling of urban traffic emission

- Developed machine learning solutions to urban air quality problems
- Carried out low-cost sensor measurements
- Performed $PM_{2.5}$ calibration and data analysis of multilevel measurements

Skills

Computer	Computational program: Matlab, Python, R, git GIS processing program: IDRISI, ArcGIS, QGIS
Language	Good written and spoken English, Native user of Chinese (Cantonese and Mandarin), Beginner of Finnish

Featured Publication

1. **Fung, P. L.**, Al-Jaghbeer, O., Chen, J., Ville-Veikko P., Vosough, S., Roncoli, C., & Järvi, L. (2025) A geospatial approach for dynamic on-road emission through open-access floating car data, *Environmental Research Letters*.
2. **Fung, P. L.**, Savadkoohi, M., Zaidan, M. A., Niemi, J. V., Timonen, H., Pandolfi, M., Alastuey, A, Querol, X., Hussein, T., & Petäjä, T. (2024) Constructing transferable and interpretable machine learning models for black carbon concentrations, *Environment International.* 184, 108449.
3. **Fung, P. L.**, Rannik, Ü., Mammarella, I., & Vesala T. (2023) Ozone Fluxes Over a Boreal Lake Exhibit Higher Deposition at Nights. *Geophysical Research Letters.* 50(*23*), e2023GL104354.
4. **Fung, P. L.**, Al-Jaghbeer, O., Pirjola, L., Aaltonen, H., & Järvi, L. (2023). Exploring the discrepancy between top-down and bottom-up approaches of fine spatio-temporal vehicular CO_2 emission in an urban road network, *Science of The Total Environment.* 901C, 165827.
5. **Fung, P. L.**, Sillanpää, S., Niemi, J. V., Kousa, A., Timonen, H., Zaidan, M. A., Saukko, E., Kulmala, M., Petäjä, T., & Hussein, T. (2022). Improving the current air quality index with new particulate indicators using a robust statistical approach, *Science of The Total Environment.* 844, 157099.
6. **Fung, P. L.**, Zaidan, M. A., Niemi, J. V., Saukko, E., Timonen, H., Kousa, A., Kuula, J., Rönkkö, T., Karppinen, A., Tarkoma, S., Kulmala, M., Petäjä, T., & Hussein, T. (2022). Input-adaptive linear mixed-effects model for estimating alveolar lung-deposited surface area (LDSA) using multipollutant datasets, *Atmospheric Chemistry and Physics.* 22, 1861–1882.
7. **Fung, P. L.**, Zaidan, M. A., Surakhi, O., Tarkoma, S., Petäjä, T., & Hussein, T. (2021). Data imputation in in situ-measured particle size distributions by means of neural networks. *Atmospheric Measurement Technique*. 14, 5535–5554.
8. **Fung, P. L.**, Zaidan, M. A., Timonen, H., Niemi, J. V., Kousa, A., Kuula, J., ... & Hussein, T. (2021). Evaluation of white-box versus black-box machine learning models in estimating ambient black carbon concentration. *Journal of Aerosol Science*. 105694.